Gyeongrok Oh

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Homepage

Research Interests

- **Multi-modal Representation Learning**: unified modeling of language, vision, and audio in large-scale foundation models.
- **Embodied AI Systems**: 3D perception and domain adaptation for real-world embodied agents.

Education

Korea University — *Seoul, South Korea*
Ph.D. in Artificial Intelligence — Sep. 2022 – Aug. 2027 (Expected)
GPA: 4.28 / 4.5
Advisor: Prof. Sangpil Kim

Inha University — *Incheon, South Korea*
B.S. in Computer Science — Mar. 2016 – Aug. 2022
GPA: 4.0 / 4.5
Advisor: Prof. Jinman Jung

Research Experience

University of Michigan — *Ann Arbor, MI, USA*
Visiting Researcher — *Aug. 2026 – Aug. 2027 (Expected)*
Advisors: Prof. Honglak Lee
Projects: World Model, Vision-Langauage-Action Model

Samsung Advanced Institute of Technology (SAIT) — *Suwon, South Korea*
Visiting Researcher, Computer Vision Lab — *Jan. 2024 – Feb. 2025*
Advisors: Ph.D. Sujin Jang
Projects: Autonomous Driving, Vision-Language Navigation

Publications († Equal contribution)

Conference Papers

[C9] Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation
Gyeongrok Oh, Youngdong Jang, Jonghyun Choi, Suk-Ju Kang, Guang Lin, Sangpil Kim
Submission, 2026 [paper]

[C8] Motion Cues from Image-based Point Tracking for LiDAR Scene Flow Estimation
Gyeongrok Oh†, Youngdong Jang†, Jong Wook Kim, Hyunju Ryu, Hyung-gun Chi, SeungHyeon Kim, Seungryong Kim, Jonghyun Choi, Sangpil Kim
NeurIPS, 2026 [paper]

[C7] Active Test-time Vision-Language Navigation
Heeju Ko, Sungjune Kim, **Gyeongrok Oh**, Jeongyoon Yoon, Honglak Lee, Sujin Jang, Seungryong Kim, Sangpil Kim
NeurIPS, 2025 [paper]

[C6] Test-time Adaptation for Online Vision-Language Navigation with Feedback-based Reinforcement Learning
Gyeongrok Oh†, Sungjune Kim†, Heeju Ko, Daehyun Ji, Dongwook Lee, Byung-Jun Lee, Sujin Jang*, Sangpil Kim*
ICML, 2025 [paper]

[C5] 3D Occupancy Prediction with Low-Resolution Queries via Prototype-aware View Transformation
Gyeongrok Oh†, Sungjune Kim†, Heeju Ko, Hyung-gun Chi, Jinkyu Kim, Dongwook Lee, Daehyun Ji, Sungjoon Choi, Sujin Jang*, Sangpil Kim*
CVPR, 2025 [paper]

[C4] MEVG: Multi-event Video Generation with Text-to-Video Models
Gyeongrok Oh, Jaehwan Jeong, Sieun Kim, Wonmin Byeon, Jinkyu Kim, Sungwoong Kim, Sangpil Kim
ECCV, 2024 [paper]

[C3] CMDA: Cross-Modal and Domain Adversarial Adaptation for LiDAR-based 3D Object Detection
Gyusam Chang†, Wonseok Roh†, Sujin Jang, Dongwook Lee, Daehyun Ji, **Gyeongrok Oh**, Jinsun Park, Jinkyu Kim*, Sangpil Kim*
AAAI, 2024 [paper]

[C2] Functional Hand Type Prior for 3D Hand Pose Estimation and Action Recognition from Egocentric View Monocular Videos
Wonseok Roh, Seung hyun Lee, Won Jeong Ryoo, **Gyeongrok Oh**, Jakyung Lee, Soo Yeon Hwang, Hyung-gun Chi, Sangpil Kim
BMVC (Oral), 2023 [paper]

[C1] Sound-Guided Semantic Video Generation
Seung hyun Lee, **Gyeongrok Oh**, Wonmin Byeon, Chanyoung Kim, Won Jeong Ryoo, Sang Ho Yoon, Jihyun Bae, Jinkyu Kim*, Sangpil Kim*
ECCV, 2022 [paper]

Journal Papers

[J4] LVMark: Robust Watermark for Latent Video Diffusion Models
Youngdong Jang*, MinHyuk Jang*, JaeHyeok Lee, Feng Yang, **Gyeongrok Oh**, Jongheon Jeong, Sangpil Kim
IEEE Transactions on Information Forensics & Security, 2026. ***IF: 8.0*** [paper]

[J3] FPANet: Frequency-based Video Demoireing using Frame-level Post Alignment
Gyeongrok Oh, Sungjune Kim, Heon Gu, Sang Ho Yoon, Jinkyu Kim, Sangpil Kim
Neural Networks, 2025. ***IF: 6.3*** [paper]

[J2] Robust Sound-Guided Image Manipulation
Seunghyun Lee†, Hyung-gun Chi†, **Gyeongrok Oh**, Wonmin Byeon, Sang Ho Yoon, Hyunje Park, Wonjun Cho, Jinkyu Kim*, Sangpil Kim*
Neural Networks, 2024. ***IF: 6.3*** [paper]

[J1] Audio-Guided Implicit Neural Representation for Local Image Stylization
Seung Hyun Lee†, Sieun Kim†, Wonmin Byeon, **Gyeongrok Oh**, Sumin In, Hyeongcheol Park, Sang Ho Yoon, Sung-hee Hong, Jinkyu Kim*, Sangpil Kim*
Computational Visual Media, 2024. ***IF: 18.3*** [paper]

Patents

[P1] Method and Apparatus with 3D Occupancy Prediction Learning US Patent
Gyeongrok Oh, Sujin Jang, Sangpil Kim, Sungjune Kim, Jinkyu Kim, Dongwook Lee, Dae Hyun Ji
US20250356594A1, 2025 (Pending)

Teaching Experience

Korea University *Seoul, South Korea*
Teaching Assistant

- Statistics (2023), Computer Vision (2025), Intelligent Mobility (2025)

LG CNS *Seoul, South Korea*
Teaching Assistant

- Anomaly Detection Course for Executives and Employees (2022, 2023)

Inha University *Incheon, South Korea*
Teaching Assistant

- Data Structures (2021)

Awards & Honors

BK21 Academic Excellence Scholarship Korea University

Departmental Scholarship Inha University
Department of Computer Science 2020–2022

Academic Service

Conference Reviewing

CVPR	2024–2026	ICLR	2027
ICCV	2025	WACV	2025–2027
ECCV	2026	AAAI	2025–2027
NeurIPS	2026	BMVC	2026

Journal Reviewing

TPAMI · T-IFS · Neural Networks · Pattern Recognition